

M624 HOMEWORK – SPRING 2026

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SET 1 - DUE 02/17/2026

From Chapter 4 :

Additional Problem 1: Prove the Auxiliary Lemma we stated in Lecture 2. Namely prove the following:

A normed vector space $(X, \|\cdot\|)$ is *complete* if and only if every absolutely convergent series in X converges in X .

Recall; i) A series $\{x_n\}_n$ is said to be absolutely convergent if $\sum_{n=1}^{\infty} \|x_n\| < \infty$.

ii) A series $\sum_{n=1}^{\infty} x_n$ converges in X if there exists an x in X such that $\lim_{N \rightarrow \infty} \sum_{n=1}^N x_n = x$ in X ; that is if $\|\sum_{n=1}^N x_n - x\| \rightarrow 0$ as $N \rightarrow \infty$

Additional Problem 2: For any $1 \leq p < \infty$ consider the space

$$L^p(\mathbb{R}^d) := \{f : \mathbb{R}^d \rightarrow \mathbb{C}, \text{ measurable, } \|f\|_{L^p(\mathbb{R}^d)} := \left(\int_{\mathbb{R}^d} |f(x)|^p dm \right)^{\frac{1}{p}} < \infty\}.$$

Assume that $\|f\|_{L^p(\mathbb{R}^d)}$ is a norm[†] whence $d_p(f, g) := \|f - g\|_{L^p(\mathbb{R}^d)}$ defines a metric and L^p is a metric space. Use the proof of completeness that I gave in class from Folland (Lecture 3) when $p = 2$ to **prove** that $L^p(\mathbb{R}^d)$ is *complete*.

(†) Question: can you guess what would you need to prove the triangle inequality when $p \neq 2$? Justify your answer.

From Chapter 4 (pp 193-194): 1, 2*, 3, 4 (show completeness only), 5, 6, 7.

* to show that $f - g$ is *orthogonal* to g you need to show that $\langle f - g, g \rangle = 0$.